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TAQIYA EHSAN (she/her)

Research: Human-AI Interaction, Causal Reasoning, Multimodal Sensing

I study human-AI interaction in physical spaces through the lens of causal AI and multimodal sensing, with applications to smart environments and human-centered autonomous systems.

EDUCATION

Rutgers University — New Brunswick, NJ
PhD in Machine Learning — 2023–present
Advisor: Dr. Jorge Ortiz, Dept. of Electrical & Computer Engineering

Rutgers University — New Brunswick, NJ
M.S. in Electrical & Computer Engineering (ML track), GPA: 4.0/4.0 — 2023–2025
Certifications: Graduate Certificate in Socially Cognizant Robotics

Rutgers University – Honors College — New Brunswick, NJ
B.S. in Electrical & Computer Engineering, GPA: 4.0/4.0 — 2019–2023
Senior Capstone: Data Insights Application for Obesity (Advisor: Dr. Sasan Haghani)

PUBLICATIONS AND POSTERS

[1] **T. Ehsan**, S. Xia, and J. Ortiz, "Act, validate, adapt: Closing the causal discovery-control loop", in *Advances in Neural Information Processing Systems*, 2026.

[2] **T. Ehsan**, S. Xia, and J. Ortiz, "Policygrid: Causal discovery for adaptive policy optimization in embodied agents (student abstract)", *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 48, pp. 41 203–41 205, Mar. 2026. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/42214.

[3] S. Xia, Q. Li, **T. Ehsan**, and J. Ortiz, "Tracefix: Repairing agent coordination protocols with tla+ counterexamples", in *Proceedings of the ACM Conference on AI and Agentic Systems*, ser. CAIS '26, Association for Computing Machinery, 2026, pp. 181–196, ISBN: 9798400724152. [Online]. Available: https://doi.org/10.1145/3786335.3813159, ***Best Paper Award***.

[4] **T. Ehsan**, S. Xia, and J. Ortiz, *Grid: Graph-based reasoning for intervention and discovery in built environments*, 2025. arXiv: 2509.16397 [cs.LG]. [Online]. Available: https://arxiv.org/abs/2509.16397.

[5] **T. Ehsan**, S. Xia, and J. Ortiz, "PolicyGRID: Acting to understand, understanding to act", in *NeurIPS 2025 Workshop on Embodied World Models for Decision Making*, 2025. [Online]. Available: https://openreview.net/forum?id=SqaOyB89rE.

[6] N. S. Pargoo, M. Ghasemi, S. Xia, M. K. Turkcan, **T. Ehsan**, C. Zang, Y. Sun, J. Ghaderi, G. Zussman, Z. Kostic, and J. Ortiz, "Urban sensing for human-centered systems: A modular edge framework for real-time interaction", in Proc. 3rd Int. Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys), 2025., 2025. [Online]. Available: https://par.nsf.gov/biblio/10582246.

[7] N. S. Pargoo, M. Ghasemi, S. Xia, M. K. Turkcan, **T. Ehsan**, C. Zang, Y. Sun, J. Ghaderi, G. Zussman, Z. Kostic, and J. Ortiz, "The streetscape application services stack (sass): Towards a distributed sensing architecture for urban applications", *arXiv preprint arXiv:2411.19714*, 2024. [Online]. Available: https://arxiv.org/pdf/2411.19714.

[8] Y. Sun, N. S. Pargoo, **T. Ehsan**, Z. Zhang, and J. Ortiz, "Vchar: Variance-driven complex human activity recognition framework with generative representation", *arXiv preprint arXiv:2407.03291*, 2024. [Online]. Available: https://arxiv.org/abs/2407.03291.

[9] **T. Ehsan**, H. Senthilkumar, A. Uddin, S. Bansal, J. Canevari, S. Motlani, and S. Haghani, "Data insights application for obesity", in *Poster Session of 45th Annual International Conference*, IEEE Engineering in Medicine & Biology Society, 2023. [Online]. Available: https://embc23-c10000.eorganiser.com.au/data/clients/1/697/submissions/164739/abstract.pdf.

[10] **T. Ehsan**, H. Senthilkumar, A. Uddin, S. Bansal, S. Motlani, J. Canevari, and S. Haghani, "Comorbiviz: A tool for visualizing comorbidity risks and costs associated with obesity", in *International Conference on e-Health and Bioengineering*, Springer, 2023, pp. 536–545. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-031-62502-2_62.

[11] T. Chowdhury, A. Bhatti, I. Mandel, **T. Ehsan**, W. Ju, and J. Ortiz, "Towards sensing urban-scale covid-19 policy compliance in new york city", in *Proceedings of the 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, ser. BuildSys '21, Coimbra, Portugal: Association for Computing Machinery, 2021, pp. 353–356, ISBN: 9781450391146. [Online]. Available: https://doi.org/10.1145/3486611.3491123.

Projects

Human-AI Interaction in Physical Spaces

Machine Causation
NIH BRAIN Initiative — May 2024 –present

- Building an end-to-end causal discovery pipeline that combines statistical methods (PC algorithm, SAM) with LLM-based domain reasoning to generate, test, and iteratively validate causal hypotheses in complex physical systems.
- Designing an intervention-based framework that selects and executes targeted experiments to efficiently recover ground-truth causal structure, reducing unnecessary edge evaluations in high-dimensional graphs.

BBQS-CAMERA
NIH BRAIN Initiative — January 2024 –present

- Developing a multimodal hardware/software framework to infer anxiety state and memory efficiency from behavioral biomarkers captured in naturalistic physical environments.
- Modeling brain–behavior relationships by fusing physiological signals, neural activity, environmental sensor streams, and Ecological Momentary Assessments (EMAs) to enable real-time, in-situ human state estimation.

Enhanced Visual Awareness for Socially Cognizant Robot
SOCRATES (NSF-NRT) — September 2024 –December 2024

- Built a real-time multi-object tracking system combining YOLOv8 detection, OC-SORT, and Kalman filtering with appearance-based re-identification, improving robustness to occlusion and misdetection in crowded campus environments.
- Deployed the tracking system on a campus guide robot via ROS2, enabling synchronized RGB-D processing and modular integration of perception and control for human-aware navigation.

AI Agents for Smart Streetscapes
NSF Center for Smart Streetscapes (CS3) — September 2023 –December 2023

- Developed a cross-modal inference pipeline to estimate pedestrian accelerometer readings from crosswalk video, enabling non-invasive behavioral sensing in urban physical spaces.
- Contributed to dataset collection and model development for visual-IMU fusion, supporting downstream human activity recognition at the city scale.

Other Projects

SAGE: Skeptical Autonomous aGent for Experimentation

MLSS 2026 Hackathon: Agentic AI for Scientific Discovery — Summer 2026

- Built an autonomous code-editing ML research agent on top of Karpathy's autoresearch loop, adding a coherence gate (culls broken proposals before any evaluation is spent) and a skeptic gate (re-tests over multiple seeds; accepts only if the gain clears the noise band) to the propose–evaluate–keep/discard cycle.
- Evaluated on MLRC-Bench Machine Unlearning (CIFAR-10 ResNet-18) and three local tasks (FashionMNIST, MAGIC Gamma Telescope, Colored MNIST); skeptic gate reduced false-accept rates 1.6–3.4$\times$ versus a greedy baseline under noisy evaluation, with agent-written CNN reaching 88.8% accuracy on FashionMNIST (vs. 74.9% baseline). Built with Luyuan Yang (U. Oklahoma), Md Musfiqur Rahman (Purdue), and Ziwei Jiang (JHU). [code]

Profiling Communication Bottlenecks in Distributed LLaMA Training

Distributed Deep Learning (ECE, Rutgers University) — Fall 2025

- Characterized TP/CP/PP/DP configurations for LLaMA-8B training on NERSC Perlmutter across 9 parallelism strategies using 16 GPUs; found that balanced TP=4/PP=4 achieved 30 TFLOP/s/GPU throughput — 2.6$\times$ faster than pure pipeline parallelism — by maximizing intra-node NVLink utilization while minimizing pipeline bubbles.
- Built custom NCCL monitoring infrastructure for Megatron-LM using probabilistic sampling and async CUDA events to profile inter-stage bandwidth and latency with $<$0.5% runtime overhead.

Data Insights Application for Obesity

Senior Capstone Project (in collaboration with Novo Nordisk) — January –May 2023

- Developed an iOS application interfacing a LightGBM comorbidity risk model with state- and national-level healthcare data, enabling pharmaceutical representatives to quantify patient risk and treatment cost at a glance.
- Conducted user experience research with Novo Nordisk data scientists to align model outputs with real-world payer negotiation workflows.

Visual Prompting for Depth Estimation

MIT CSAIL — Summer –Fall 2022

- Investigated visual prompting as a parameter-efficient adaptation strategy for depth estimation, fine-tuning prompts on top of frozen CNN and Vision Transformer backbones across multiple datasets.
- Designed experiments demonstrating out-of-distribution generalization improvements, outperforming state-of-the-art baselines on held-out benchmark metrics.

Analyzing Social Distancing Based on Sensory Inputs

Rutgers WINLAB — Summer 2021

- Processed a corpus of over 300 million dashboard and traffic camera images from Nexar using YOLO-based detection and a custom image quality filter (NumPy norm + geometric mean brightness thresholding) to isolate valid pedestrian frames.
- Designed a multi-step preprocessing pipeline feeding a distance estimation algorithm to assess COVID-19 social distancing compliance across New York City.

Research & Work Experience

Rutgers Sensing & Reasoning Lab — Piscataway, NJ

Graduate Research Assistant — 2023 –Present

- Building a causal discovery pipeline combining PC algorithm, SAM, and LLM-based reasoning for hypothesis-driven intervention testing in complex physical systems (NIH BRAIN Initiative) — see Projects.
- Leading the BBQS-CAMERA project: developing multimodal models to infer anxiety and memory efficiency from behavioral biomarkers captured in naturalistic environments (NIH BRAIN Initiative) — see Projects.

Canfield Scientific, Inc. Parsippany, NJ
Computer Science Intern June –August 2025

- Designed and implemented a new CV/ML pipeline for skin texture and pore detection, improving accuracy and repeatability for clinical deployment in VISIA® and NEXA® imaging systems.
- Contributed to R&D patent documentation supporting the protection of novel skin imaging technologies.

CSAIL, Massachusetts Institute of Technology Cambridge, MA
Research Intern June –December 2022

- Led an independent research project on visual prompting for depth estimation in collaboration with a PhD student under direct faculty supervision; produced a research proposal, poster, and technical report — see Projects.

Rutgers University — CyberPhysical Intelligence Lab New Brunswick, NJ
Undergraduate Research Assistant September –December 2021

- Collaborated with a PhD student to build an ambient acoustic scene classifier, designing the feature extraction pipeline and evaluating dataset candidates for household activity recognition.

Rutgers WINLAB Piscataway, NJ
Summer Research Intern June –August 2021

- Processed 300M+ NYC traffic camera images to assess COVID-19 social distancing compliance, implementing a YOLO-based detection and image quality filtering pipeline on a remote server — see Projects.

Teaching & Mentoring

- **Workshop Facilitator & Academic Skills Instructor**, Rutgers Learning Centers Fall 2024 –Present
  Design and deliver workshops and a 4-week course (Back on Track) on metacognition, time management, procrastination, and exam preparation; provide individual and group academic coaching.
- **Guest Speaker**, Multimodal ML for Sensing Systems (ECE579), Rutgers University October 2026
  Lecture on closing the causal-discovery control loop in embodied AI.
- **Summer Research Mentor**, Wireless Information Network Laboratory (WINLAB) Summer 2026
  Coordinated and mentored undergraduate and high school interns in research methodologies, development environment setup, and project troubleshooting; supported final presentations at program open house; facilitated communication between interns and faculty advisors. Project blog: From a Prompt to Verified Agent Applications
- **Guest Speaker**, Multimodal ML for Sensing Systems (ECE579), Rutgers University November 2025
  Lecture on causal reasoning for embodied AI.
- **Undergraduate Teaching Assistant**, Data Structures (CS112), Rutgers University Fall 2022 –Spring 2023
  Held office hours, graded assignments, and supported 200+ students in foundational CS concepts.
  Supervisor: Dr. Ana Paula Centeno.
- **Learning Assistant**, Honors College Forum (SAS125), Rutgers University Fall 2020 –Spring 2022
  Facilitated discussion sections and supported first-year students in academic transition.
  Supervisor: Dr. Talia Robbins.

Scholarships and Awards

- Graduate Rapid Fire Finalist, Society of Women Engineers Annual Conference November 2026
  (6.6% acceptance of 152 submitted abstracts)
- Paul Panayotatos Endowed Scholarship 2025–2026
- School of Engineering Fellowship 2025–2026
- ECE Leadership and Service Award 2024–2025
- Rutgers Eugene V. DuBois Fellowship 2023–2024
- Matthew Leydt Society Inductee 2023

- Rutgers James Dickson Clark Scholarship 2019–2023
- Dean's List 2019–2023

Extracurricular Activities

- **Academic Coach**, Rutgers Learning Centers 2024–Present
  Provide individual and group coaching on study strategies, time management, and exam preparation.
- **Participant**, Academic Leadership for Women in Engineering (ALWE), Society of Women Engineers 2025–2026
  Year-long leadership program for women engineering academics (cohort of 50); workshops, coaching, and capstone project.
- **Member**, Doctoral Students Advisory Board 2025–2026
  Represent PhD student interests to Rutgers Career Services; advise on professional development programs and policies.
- **Founding Member & President**, ECE Graduate Students Association 2024–2026
  Founded the department's first graduate student org; lead programming for 80+ members across community, industry, and networking.
- **Assistant Residence Life Coordinator**, Honors College & Brett Hall 2023–2024
  Supervised 17 RAs across 2 buildings (~750 residents); primary on-call contact for crisis management.
- **Peer Mentor**, Society of Women Engineers 2020–2024
  Provided 1:1 mentorship to underclasswomen engineers through the SWE Leaders-Learners Program.
- **Resident Assistant**, Sojourner Truth Apartments 2021–2023
  Supported ~100 residents through conflict mediation, resource referrals, and community programming.
- **Secretary**, Society of Women Engineers 2022–2023
  Managed communications and documentation; co-organized the Women in Engineering National Conference trip.
- **First Year Mentor**, Honors College Ally 2022–2023
  Guided incoming Honors College students through academic and extracurricular transition.
- **E-board Member**, Honors College Student Advisory Board 2021–2022
  Advocated for underrepresented majors; presented student feedback and recommendations directly to Deans.